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Project Report On

"Automatic distracted driver detection using deep convolutional neural networks"

Submitted by

Vivek Kumar [20203003]

University Institute of Technology, University of Burdwan

DEPARTMENT OF INFORMATION TECHNOLOGY

BACHELOR OF ENGINEERING (B.E.)

CERTIFICATE OF APPROVAL

This is to certify that the project entitled "**Automatic driver distraction detection using deep convolutional neural networks**" submitted by **Vivek Kumar [20203003]** under the guidance and supervision **Prof. (Dr.) Shiladitya Pujari** as partial fulfilment for the award of the degree of **BACHELOR OF ENGINEERING** in **INFORMATION TECHNOLOGY at UNIVERSITY INSTITUTE OF TECHNOLOGY**. The University of Burdwan is a record of the work of the students which has been carried out under any supervision.
I hereby forward this project.

Date:…………..

Place :………….

(Project Guide)
Prof. (Dr.) Shiladitya Pujari

Assistant Professor,

DEPARTMENT OF INFORMATION TECHNOLOGY
UNIVERSITY INSTITUTE OF TECHNOLOGY

University Institute of Technology, University of Burdwan

CERTIFICATE

This is to certify that the work embodied in the final year project "**Automatic driver distraction detection using deep convolutional neural networks"** submitted by **Vivek Kumar** to the Department of Information Technology are carried out under my direct Supervision and guidance. The project work has been prepared as per regulation of The University of Burdwan and I strongly recommend that this project work can be accepted in partial fulfillment.

Date.............

Date...............

Head of the Department :
Mrs. Kasturi Ghosh
Professor & In-charge, Dept. of IT
University Institute of Technology

Project Guider :
Prof. (Dr.) Shiladitya Pujari
Lecturer, Dept. of IT
University Institute of Technology

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Date…………….

Vivek Kumar

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DECLARATION

We clarify that:

1. The work contained in the thesis is original and has been done by ourselves under the general supervision of our supervisor(s).

2. The work has not been submitted to any other Institute for any degree or diploma.

3. We have followed the guidelines provided by the Institute inwriting the thesis.

4. We have conformed to the norms and guidelines given in theoretical code of conduct of the Institute.

5. Whenever we have used materials (data, theoretical analysis, and text) from other sources, we have given due credit to them by citing them in the text of the thesis and giving their details in the references.

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Abstract

Recently, the number of road accidents has been increased worldwide due to the distraction of the drivers. This rapid road crush often leads to injuries, loss of properties, even deaths of the people. Therefore, it is essential to monitor and analyze the driver's behavior during the driving time to detect the distraction and mitigate the number of road accident. To detect various kinds of behavior like- using cell phone, talking to others, eating, sleeping or lack of concentration during driving; machine learning/deep learning can play significant role. However, this process may need high computational capacity to train the model by huge number of training dataset. In this paper, we made an effort to develop CNN based method to detect distracted driver and identify the cause of distractions like talking, sleeping or eating by means of face and hand localization. Four architectures namely CNN, VGG-16, ResNet50 and MobileNetV2 have been adopted for transfer learning. To verify the effectiveness, the proposed model is trained with thousands of images from a publicly available dataset containing ten different postures or conditions of a distracted driver and analyzed the results using various performance metrics. The performance results showed that the pre-trained MobileNetV2 model has the best classification efficiency.

objective

Driver distraction detection systems aim to enhance road safety by addressing the critical issue of distracted driving. With the increasing use of in-vehicle technology and mobile devices, driver distraction has become a significant cause of road accidents. The main objectives of driver distraction detection can be elaborated as follows:

1.Reducing Road Accidents

The primary goal is to reduce the number of road accidents caused by driver distraction. Distracted driving is a leading cause of traffic accidents, resulting in injuries and fatalities. By detecting distraction early, these systems can help prevent accidents and save lives.

2. Enhancing Overall Road Safety

Driver distraction detection systems contribute to overall road safety. By ensuring that drivers remain focused on the road, these systems help create a safer driving environment for everyone. This includes not only the drivers and passengers in the vehicle but also pedestrians and other road users

3. Providing Real-Time Warnings and Alerts

One of the critical functions of these systems is to provide real-time feedback to drivers. When distraction is detected, the system can issue alerts or warnings to refocus the driver's attention on the road. These alerts can be visual, auditory, or haptic (e.g., seat vibrations), depending on the system design.

4. Utilizing Advanced Detection Technologies

Driver distraction detection systems employ various advanced technologies such as cameras, sensors, and artificial intelligence (AI) algorithms. These technologies monitor driver behavior, including eye movements, head position, and hand movements, to identify signs of distraction. Machine learning models can analyze this data to distinguish between attentive and distracted states.

5. Supporting Autonomous and Semi-Autonomous Driving

As the automotive industry moves towards autonomous and semi-autonomous vehicles, driver distraction detection systems become even more critical. In semi-autonomous vehicles, the driver still needs to be ready to take control when necessary. These systems ensure that drivers remain attentive and can take over when required, thereby supporting the transition to fully autonomous driving.

6. Encouraging Safe Driving Practices

Driver distraction detection systems also play a role in encouraging safe driving practices. By providing continuous feedback and alerts, they remind drivers of the importance of maintaining focus on driving. Over time, this can help cultivate safer driving habits and reduce reliance on potentially distracting activities while driving.

7. Data Collection and Analysis

Driver distraction detection systems collect valuable data on driver behavior and distraction patterns. This data can be analyzed to understand the common causes and contexts of distraction. Insights gained from this analysis can inform the design of better driver assistance systems and influence traffic safety policies.

8. Integration with Other Safety Systems

These systems can be integrated with other in-vehicle safety systems, such as lane-keeping assist, adaptive cruise control, and emergency braking systems. By working together, these systems provide a comprehensive approach to vehicle safety, addressing various factors that can lead to accidents

In summary, the objective of driver distraction detection systems is multifaceted, aiming to enhance road safety, reduce accidents, provide real-time alerts, utilize advanced detection technologies, support autonomous driving, collect and analyze data, integrate with other safety systems, and encourage safe driving practices. By achieving these objectives, these systems can significantly contribute to a safer driving environment, ultimately saving lives and reducing the burden of road accidents on society.

Motivation

Both driver inattention and driver distraction are huge challenges in road traffic, resulting in a high number of accidents and fatalities every year. For this reason, vehicle manufacturers, suppliers, start-ups, and researchers are devoting more and more resources to better understand and measure the causes of driver distraction and inattention. Thereby, they develop warning and prevention mechanisms for drivers or increase the automation level of the vehicle to avoid dealing with driver distraction in the first place, e.g., with automated driving functionalities. Some modern vehicles above a certain vehicle class and equipment level may already have simple systems that can detect certain types of driver inattention, such as driver fatigue and warn the driver accordingly. Such systems are usually subsumed under the term driver assistance systems, which also includes vehicle automation systems such as adaptive distance keeping or lane keeping. With the increasing automation of the driving function, more and more powerful systems will find their way into vehicles, with the aim of fully automated or autonomous driving. Consequently, there is also increasing research and publication on driver distraction detection and related topics. To increase understanding of driver distraction methods, we present in the paper the results of a scientific literature review to identify existing driver distraction detection methods, carefully analyze them, and extract the information flow for distraction detection. In the process, the identified methods are transferred into a framework for the detection of distracted driving to capture and structure the entire spectrum of distraction detection

Introduction

The number of cars sold yearly exceeds 70 million, and there are more than 1.3 million fatal vehicle accidents yearly. India is responsible for 11% of all traffic-related fatalities worldwide. 78% of accidents are attributed to drivers. It is heartbreaking to see that road accidents are a leading cause of death for young people between the ages of 5 and 29 according to the WHO report. Tragically, the number of fatalities keeps increasing each year due to driver distraction. In India, 17 people lose their lives every hour due to vehicle accidents. Driver distraction is the most common cause of motor vehicle accidents and is defined as any activity that diverts the driver's attention away from their primary task. The driver is the most crucial component of the vehicle's control system, which includes steering, braking, acceleration, and additional processes. All traffic participants must be able to complete these essential activities safely. This research suggests a method for detecting driver distractions that recognizes various forms of distractions by watching the driver through a camera. Our objective is to create a high accuracy system that can track the driver's movement in real- time and determine if the driver is operating the vehicle safely or engaging in a certain type of distraction. The system will classify them appropriately using adequate Machine Learning based on their activities

Literature Survey

The reviews of some of the pertinent and important work from the literature for detecting distracted driving are summarized in this section. The major cause of manual distractions is the usage of cell phones.

Researchers employed a support vector machine (SVM)-based model to identify cell phone use while driving. The dataset used for the model focused on two driving activities: a motorist with a phone and a driver without a phone, and depicted both hands and faces with a predefined assumption. Frontal photographs of the drivers were utilised with an SVM classifier to detect the driver's behaviour. Subsequently, other researchers studied the detection of cell phone use while driving. They employed a camera mounted above the dashboard to create a database and a Hidden Conditional Random Fields model to identify cell phone usage. Zhang et al. mainly utilised hand, mouth, and face features for this purpose. In 2015, Nikhil et al. developed a dataset for hand detection in the automotive environment and utilised an Aggregate Channel Features (ACF) object detector to achieve an average precision of 70.09%.

Driver distractions can be detected using a variety of visual indicators and mathematical models. This study examines the use of machine learning (ML) algorithms to identify driver distractions. ML models are trained to recognize certain patterns associated with distracted driving, such as pupil diameter, eye gaze, head posture, facial expressions, and driving posture. These models are then used to identify driver distractions in real time.

A support vector machine-based (SVM-based) model was created to identify drivers using cell phones while operating a vehicle by extracting features from an image. The driver's face was depicted in frontal view images for the dataset, with and without a phone.

Researchers have been developing datasets and object detectors to detect hands in an automotive environment. one study, an object detector with aggregate channel features was used to achieve an average precision of 70.09%. Another study focused on detecting a driver using a cell phone. The authors used AdaBoost classifiers and a histogram of gradients (HOGs) to locate the landmarks on the face, and then extracted bounding boxes from the left to the right side of the face. Segmentation and training were used to achieve 93.9% accuracy at 7.5 frames per second. To further explore distracted driving, a more comprehensive dataset was created that took into **account four** different activities: safe driving, using the shift lever, eating, and talking on a cell phone. The contourlet transform and random forest were used by the authors to achieve an accuracy of 90.5%. Additionally, Faster R-CNN was suggested and a system with a pyramid of the histogram of gradients (PHOG) and multilayer perceptron was used to produce an accuracy of 94.75%,

In recent years, the State Farm distracted driver identification competition on Kaggle has become the first publicly available dataset to consider a wide range of distractions. This dataset outlines ten postures to be detected, including safe driving and nine distracting behaviours. To accurately identify these postures, many researchers have turned to a combination of handcrafted feature extractors, such as SIFT, SURF, and HoG, and classical classifiers like SVM, BoW, and NN. It is clear, however, that CNN's have proven to be the most successful technique for achieving high accuracy. Although the dataset is available for public use, the rules and regulations of the competition restrict its use to competition purposes only

Methodology

CNNs have made significant strides in recent years in a number of applications, including image classification, object identification, action recognition, natural language processing. and many more. Convolutional filters/layers, Activation functions, Pooling layer, and Fully Connected (FC) layer are the fundamental components of a CNN-based system. These layers are effectively stacked on top of one another to create a CNN. Since 2012, CNNs have advanced quite quickly thanks to the availability of massive amounts of labelled data and computational capacity. In the field of computer vision, many architectures like as AlexNet, ZFNet, VGGNet, GoogLeNet, and ResNet have developed standards. In this study, we investigate and tweak the Simonyan and Zisserman [16] VGG-16 architecture for the job of distracted driving detection.

Convolutional Neural Network (CNN):

A Convolutional Neural Network (CNN) consists of an input layer, a convolution layer, a pooling layer, a fully connected layer, and an output layer (see Figure 1). The input layer takes in images of the driver's current state and the convolution layer uses these images to extract features. The pooling layer calculates the feature values from the extracted features. Depending on the complexity of the images, the convolution and pooling layer can be extended in order to extract more details. The fully connected layer combines the output from the earlier layers into a single vector, which can then be used as input for the next layer. Finally, the output layer categorises the plant disease based on the input.

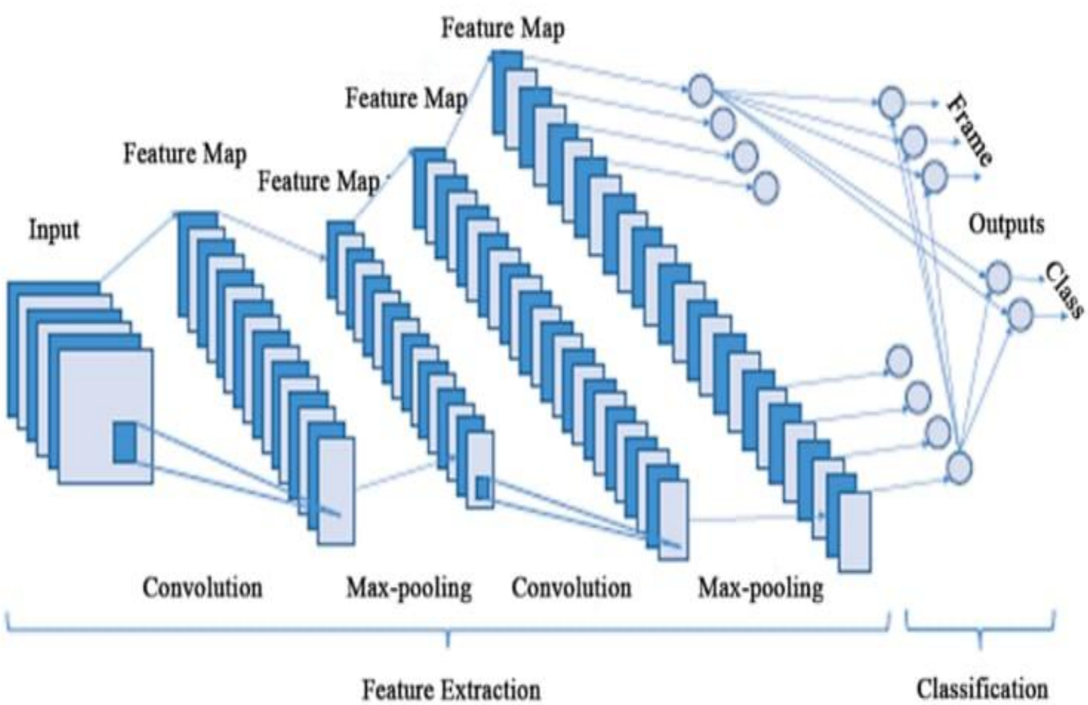

Figure:1

VGG-16 and VGG-19:

VGG is a standard deep Convolutional Neural Network (CNN) architecture consisting of multiple layers. It was developed by the Visual Geometry Group and is the basis of a large number of object recognition models. VGG-16 and VGG-19 consist of 16 and 19 convolutional layers respectively and are capable of outperforming many other models on a wide range of tasks and datasets. It remains one of the most popular architectures for image recognition

the VGGNet-16 supports 16 layers and can classify images into 1000 object categories, including keyboard, animals, pencil, mouse, etc. Additionally, the model has an image input size of 224-by-224.

The concept of the VGG19 model (also VGGNet- 19) is the same as the VGG16 except that it supports 19 layers. The "16" and "19" stand for the number of weight layers in the model (convolutional layers). This means that VGG19 has three more convolutional layers than VGG16.

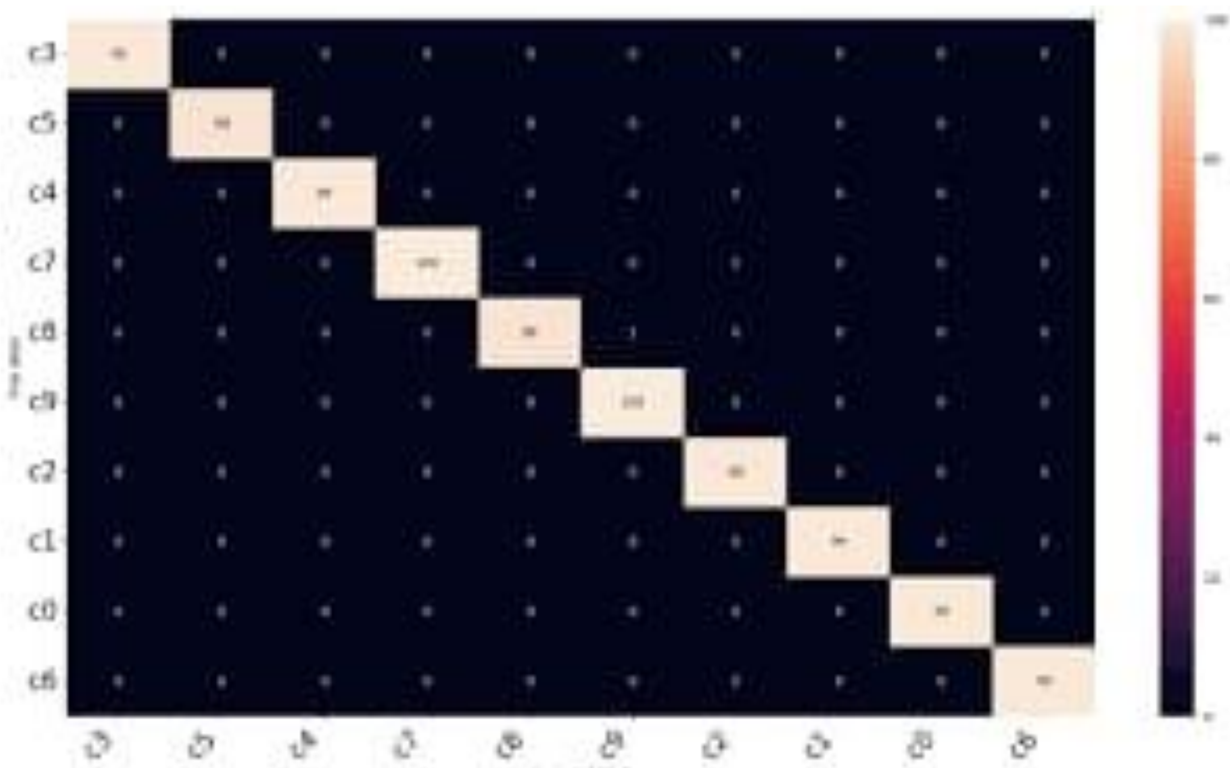

Figure:2

Data Set

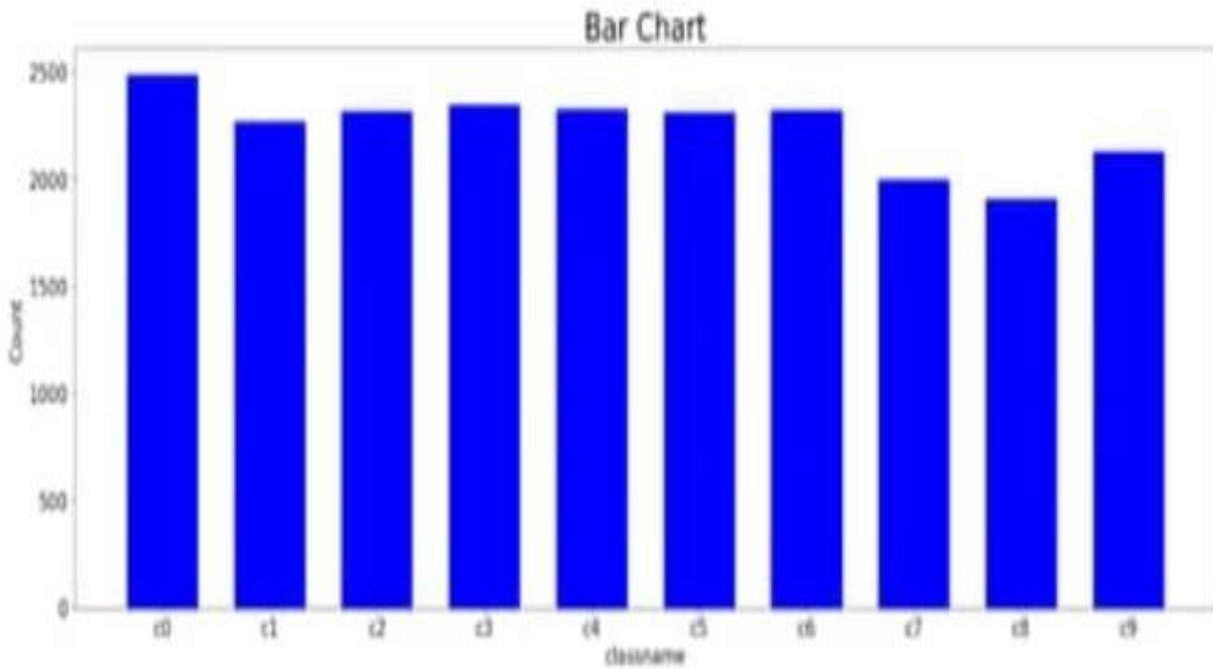

Figure:3

The dataset has collected the training images from the State Farm. It consists of a variety of images distributed in 10 categories. The distribution is arranged in a way so that there will be very less possibility of duplicate images. All the images are of a driver sitting inside the car. There are a total of 22,424 images classified into 10 categories. The count of images in every category is shown in figure [3]. The categories are:

- CO: safe driving
- C1: texting-right
- C2: talking on the phone - right
- C3: texting-left
- C4: talking on the phone – left
- CS: operating the radio
- C6: drinking
- C7: reaching behind
- C8: hair and makeup
- C9: talking to a passenger

Results

Figure:4

Figure:5

With the hope of increasing accuracy, the convolutional neural network is being tested for the detection of a distracted driver. The database is split into two datasets, training, and testing. CNN determines whether a Driver is distracted or not, and if so, it also forecasts the type of distraction. The CNN model was trained using a 25-epoch. The performance of the CNN model on the testing dataset during training is shown in Figures 4 and 5. Figure 2 displays a sample confusion matrix

Conclusions

In conclusion, driver distraction detection systems are critical for advancing road safety in today's increasingly connected driving environments. These systems address the growing problem of distracted driving, which is a leading cause of traffic accidents, injuries, and fatalities. By utilizing advanced technologies such as cameras, sensors, and AI algorithms, these systems can monitor and analyze driver behavior in real-time, detecting signs of distraction and providing immediate feedback through visual, auditory, or haptic alerts.

The primary objective of these systems is to reduce the incidence of accidents caused by distracted driving. By identifying distractions early, these systems can help prevent potential collisions and save lives. Additionally, they contribute to the overall enhancement of road safety by ensuring that drivers remain focused on the task of driving, thereby protecting not only the drivers and passengers but also other road users such as pedestrians and cyclists.

Moreover, driver distraction detection systems play a pivotal role in the evolution of autonomous and semi-autonomous vehicles. In semi-autonomous vehicles, these systems ensure that drivers are attentive and ready to take control when necessary, thus supporting a safer transition towards fully autonomous driving. The data collected by these systems also provides valuable insights into driver behavior and distraction patterns, informing the development of more effective driver assistance technologies and influencing traffic safety regulations and policies.

Furthermore, the integration of distraction detection systems with other in-vehicle safety features, such as lane-keeping assist and adaptive cruise control, offers a comprehensive approach to vehicle safety. This integration enhances the overall effectiveness of safety measures, providing a multi-faceted defense against various factors that can lead to accidents.

By continuously reminding drivers of the importance of maintaining focus and encouraging safer driving habits, these systems foster a culture of safety on the roads. Over time, this can lead to a significant reduction in distracted driving incidents and promote a more responsible driving behavior among motorists.

In essence, driver distraction detection systems are indispensable tools in the modern automotive safety landscape. They not only mitigate the risks associated with distracted driving but also support broader safety objectives, contributing to a significant reduction in road accidents and enhancing the overall safety of the driving environment. Through their implementation and continuous development, we move closer to achieving safer roads and protecting the lives of all road users.

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